ElGamal Public Key Cryptography

CS 303 Alg. Number Theory & Cryptography

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Taher ElGamal, "A Public-Key Cryptosystem and a Signature Scheme Based on Discrete Logarithms", IEEE Transactions on Information Theory, v. IT-31, n. 4, 1985, pp469–472 or CRYPTO 84, pp10–18, Springer-Verlag.

Outline

- Primitive Element Theorem
- Diffie Hellman Key Distribution
- ElGamal Encryption
- ElGamal Digital Signatures

Public Key Cryptography

- Let M be a message and let C be the encrypted message (ciphertext). A public key cryptosystem has a separate method E() for encrypting and D() decrypting.
  - D(E(M)) = M
  - Both E() and D() are easy to compute
  - Publicly revealing E() does not make it easy to determine D()
  - E(D(M)) = M - needed for signatures

- The collection of E()'s are made publicly available but the D()'s remain secret. Called a one-way trap-door function (hard to invert, but easy if you have the secret information)

Order

- Definition. Let $b \in Z_n^*$ The order of b is the smallest positive integer satisfying $b^e \equiv 1 \pmod{n}$.

- Theorem 1. If b has order e modulo n and if j is a positive integer such that $b^j \equiv 1 \pmod{n}$, then $e|j$.

Proof. $j = q \times e + r$, $0 \leq r < e$.

$b^j \equiv 1 \equiv (b^e)^q b^r \equiv b^r \pmod{n}$.

This implies that r = 0, since e is the smallest power of b equivalent to 1 mod n.

- Corollary 2. Let $b \in Z_n^*$. $ord(b)|\phi(n)$.

Primitive Element Theorem

- $Z_p^* = \langle\alpha\rangle$, i.e. $\text{ord}(\alpha) = p-1$.

- Example
  - $Z_7^* = \langle 3\rangle$ $3^1=3, 3^2=2, 3^3=6, 3^4=4, 3^5=5, 3^6=1$
  - $Z_{13}^* = \langle 2\rangle$ $2^1=2, 2^2=4, 2^3=8, 2^4=3, 2^5=6, 2^6=12, 2^7=11, 2^8=9, 2^9=5, 2^{10}=10, 2^{11}=7, 2^{12}=1$

- Note. $\text{ord}(\alpha) = p-1 \Rightarrow \{1, \alpha, \alpha^2, \ldots, \alpha^{p-1}\}$ distinct.

Discrete Logarithms

- Discrete log problem
  - Given $Z_p^* = \langle \alpha \rangle$
  - $\log_\alpha(y) = x$, if $y = \alpha^x$.

- Example
  - $Z_{13}^* = \langle 2 \rangle$ $2^1=2$, $2^2=4$, $2^3=8$, $2^4=3$, $2^5=6$, $2^6=12$, $2^7=11$, $2^8=9$, $2^9=5$, $2^{10}=10$, $2^{11}=7$, $2^{12}=1$
  - $\text{Log}_2(5) = 9$.

Properties of Primitive Elements

- Theorem 3. If b has order e modulo n, then ord(b^i) = e/gcd(e,i).
- Theorem 4. Let p be a prime and d a divisor of p-1, then the number of positive integers less than p with order d is $\phi(d)$.
- Corollary 5. The number of primitive elements mod p is equal to $\phi(p-1) > 1$.

Some Lemmas

- Lemma 6. Let P(x) be a polynomial of degree t and let p be a prime. If p does not divide the coefficient of x^t in P(x), then $P(x) \equiv 0 \pmod{p}$, has at most t solutions mod p.

Proof. By induction on the degree of P(x)=t.

$P(x_1) = 0 \Rightarrow P(x) = P_1(x) \times (x - x_1)$, and the degree of $P_1(x)$ = t-1.

- Lemma 7. The sum of $\phi(d)$ over the divisors of n = n.
  - Example: n=12. $\phi(1)+ \phi(2)+ \phi(3)+ \phi(4)+ \phi(6)+ \phi(12)=1+1+2+2+2+4 = 12$.

Primitive Element Theorem

- Theorem. Let p be a prime and d a divisor of p-1, then the number of positive integers less than p with order d is $\phi(d)$.

Proof. If there is an element a of order d, then by Theorem 3, a^i, gcd(i,d)=1 is also of order d. By Lemma 6, 1, a, a^2,…,a^{d-1} are the roots of $P(x)=x^d-1$, and there $\phi(d)$ elements of order d. Since every elements is of order d|p-1 and $p-1 = \Sigma_{d|p-1} \phi(d)$, there must be an element of order d for every d|p-1 and hence exactly $\phi(d)$ of them.

Public Key Distribution

- The goal is for two users to securely exchange a key over an insecure channel. The key is then used in a normal cryptosystem

- Diffie-Hellman Key Exchange
  - $A = g^a \bmod p$ (p prime, g primitive – all elements of $(Z_p)^*$are powers of g) [Alice sends A to Bob]
  - $a = \log_g A \bmod p$ [discrete log]
  - $B = g^b \bmod p$ [Bob sends B to Alice]
  - $K = g^{ab} \bmod p$ [shared key]
  - $A^b = g^{ab} = B^a \bmod p$

ElGamal Encryption

- $Z_p^* = \langle g \rangle$, $m \in Z_p$ message
  - B encrypts a message to A.
- Alice: a random, $h = g^a$, public key = (p, g,A)
- Bob: k random (ephemeral key), $c_1 = g^k$, shared key $K = A^k = g^{ak}$
  - $E_A(m) = (c_1,c_2)$, $c_2 = mK \bmod p$.
  - $D_A((c1,c2)) = c_2 * (1/K) \bmod p$, $K = c_1{}^a = g^{ak}$

- Security depends on Computational Diffie-Hellman (CDH) assumption: given (g, g^a, g^b) it is hard to compute g^{ab}
- Do not use same k twice

ElGamal Digital Signature

- $Z_p^* = \langle g \rangle$, $m \in Z_p$ message
  - A signs message m.

- Alice: $A = g^a$, public key = (p, g,A), secret key = x.
- Alice: k random with gcd(k,p-1)=1
  - $r = g^k \pmod p$
  - $s = (m - xr)(1/k) \bmod p-1$ [$m = sk + xr \pmod{p-1}$]
  - Signature = (r,s)
  - Verify $g^m = r^s h^r$